Create nodes and relationships in Neo4j (e.g., Students-Friends, Company-Employee)

Step 1: Access Neo4j (Docker)

docker exec -it neo4j-container cypher-shell -u neo4j

-p password

```
ubuntu@Suhel-PC:~$ docker exec -it neo4j-container cypher-shell -u neo4j
 -p password
WARNING: A restricted method in java.lang.System has been called
WARNING: java.lang.System::load has been called by org.fusesource.jansi.
internal.JansiLoader in an unnamed module (file:/var/lib/neo4j/lib/jansi
-2.4.0.jar)
WARNING: Use --enable-native-access=ALL-UNNAMED to avoid a warning for c
allers in this module
WARNING: Restricted methods will be blocked in a future release unless n
ative access is enabled

Connected to Neo4j using Bolt protocol version 6.0 at neo4j://localhost:
7687 as user neo4j.
Type :help for a list of available commands or :exit to exit the shell.
Note that Cypher queries must end with a semicolon.
neo4j@neo4j>
```

Step 2: Create Nodes

CREATE (s1:Student {name: 'Alice', age: 22}),

(s2:Student {name: 'Bob', age: 23}),

(s3:Student {name: 'Charlie', age: 21});

```
neo4j@neo4j> CREATE (s1:Student {name: 'Alice', age: 22}),
                    (s2:Student {name: 'Bob', age: 23}),
                    (s3:Student {name: 'Charlie', age: 21});
0 rows
ready to start consuming query after 303 ms, results consumed after anot
her 0 ms
Created 3 nodes, set 6 properties, added 3 labels
```

CREATE (c:Company {name: 'TechCorp'}),

(e1:Employee {name: 'John', role: 'Developer'}),

(e2:Employee {name: 'Emma', role: 'Manager'});

```
neo4j@neo4j> CREATE (c:Company {name: 'TechCorp'}),
                    (e1:Employee {name: 'John', role: 'Developer'}),
                    (e2:Employee {name: 'Emma', role: 'Manager'});
0 rows
ready to start consuming query after 76 ms, results consumed after anoth
er 0 ms
Created 3 nodes, set 5 properties, added 3 labels
```

Step 3: Create Relationships

MATCH (a:Student {name: 'Alice'}), (b:Student {name: 'Bob'})

CREATE (a)-[:FRIENDS_WITH]->(b);

MATCH (b:Student {name: 'Bob'}), (c:Student {name: 'Charlie'})

CREATE (b)-[:FRIENDS_WITH]->(c);

```
neo4j@neo4j> CREATE (c:Company {name: 'TechCorp'}),
                    (e1:Employee {name: 'John', role: 'Developer'}),
                    (e2:Employee {name: 'Emma', role: 'Manager'});
0 rows
ready to start consuming query after 76 ms, results consumed after anoth
er 0 ms
Created 3 nodes, set 5 properties, added 3 labels
neo4j@neo4j> MATCH (a:Student {name: 'Alice'}), (b:Student {name: 'Bob'}
)
             CREATE (a)-[:FRIENDS_WITH]->(b);

             MATCH (b:Student {name: 'Bob'}), (c:Student {name: 'Charlie
'})
             CREATE (b)-[:FRIENDS_WITH]->(c);
0 rows
ready to start consuming query after 344 ms, results consumed after anot
her 0 ms
Created 1 relationship
0 rows
ready to start consuming query after 101 ms, results consumed after anot
her 0 ms
Created 1 relationship
neo4j@neo4j>
```

MATCH (c:Company {name: 'TechCorp'}), (e1:Employee {name: 'John'})

CREATE (e1)-[:WORKS_AT]->(c);

MATCH (c:Company {name: 'TechCorp'}), (e2:Employee {name: 'Emma'})

CREATE (e2)-[:WORKS_AT]->(c);

```
neo4j@neo4j> MATCH (c:Company {name: 'TechCorp'}), (e1:Employee {name: '
John'})
             CREATE (e1)-[:WORKS_AT]->(c);

             MATCH (c:Company {name: 'TechCorp'}), (e2:Employee {name: '
Emma'})
             CREATE (e2)-[:WORKS_AT]->(c);
0 rows
ready to start consuming query after 122 ms, results consumed after anot
her 1 ms
Created 1 relationship
0 rows
ready to start consuming query after 92 ms, results consumed after anoth
er 0 ms
Created 1 relationship
```

Step 4: Visualize Graph

MATCH (n)-[r]->(m)

RETURN n, r, m;

```
neo4j@neo4j> MATCH (n)-[r]->(m)
             RETURN n, r, m;
+-----------------------------------------------------------------------
--------------------------------+
| n                                                | r               | m
                                 |
+-----------------------------------------------------------------------
--------------------------------+
| (:Student {name: "Alice", age: 22})              | [:FRIENDS_WITH] | (:St
udent {name: "Bob", age: 23})     |
| (:Student {name: "Bob", age: 23})                | [:FRIENDS_WITH] | (:St
udent {name: "Charlie", age: 21}) |
| (:Employee {name: "John", role: "Developer"}) | [:WORKS_AT]     | (:Co
mpany {name: "TechCorp"})         |
| (:Employee {name: "Emma", role: "Manager"})   | [:WORKS_AT]     | (:Co
mpany {name: "TechCorp"})         |
+-----------------------------------------------------------------------
--------------------------------+

4 rows
ready to start consuming query after 129 ms, results consumed after anot
her 23 ms
neo4j@neo4j>
```

Create Everything in one query

CREATE

(a:Student {name:'Alice'}),

(b:Student {name:'Bob'}),

(c:Student {name:'Charlie'}),

(tc:Company {name:'TechCorp'}),

(j:Employee {name:'John'}),

(e:Employee {name:'Emma'}),

(a)-[:FRIENDS_WITH]->(b),

(b)-[:FRIENDS_WITH]->(c),

(j)-[:WORKS_AT]->(tc),

(e)-[:WORKS_AT]->(tc);

```
neo4j@neo4j> CREATE
             (a:Student {name:'Alice'}),
             (b:Student {name:'Bob'}),
             (c:Student {name:'Charlie'}),
             (tc:Company {name:'TechCorp'}),
             (j:Employee {name:'John'}),
             (e:Employee {name:'Emma'}),

             (a)-[:FRIENDS_WITH]->(b),
             (b)-[:FRIENDS_WITH]->(c),
             (j)-[:WORKS_AT]->(tc),
             (e)-[:WORKS_AT]->(tc);
0 rows
ready to start consuming query after 100 ms, results consumed after anot
her 0 ms
Created 6 nodes, created 4 relationships, set 6 properties, added 6 labe
ls
neo4j@neo4j>
```